# TOPIC:Student management system

## HLD (High-Level Design)

## RAG-Based Student Management Assistant

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### 1. System Overview

This system is an AI-powered Student Management Assistant built using RAG (Retrieval-Augmented Generation). It allows users to query student-related information (like student details, course info, etc.) from stored documents or database using natural language.

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### 2. Problem Definition

Traditional student systems:

- Require manual search
- Not user-friendly
- No intelligent query system

Solution:

- Ask questions like:
  **“Show students in BCA”**
  **“Details of student Rahul”**

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## 3. Architecture Diagram

## 4. Component Description

- **Document Loader** → student data load karta hai
- **Chunking** → data ko small parts me divide
- **Embedding Model** → text → vector
- **Vector DB (ChromaDB)** → embeddings store
- **Retriever** → relevant data fetch

- **LLM** → answer generate
- **LangGraph** → flow control
- **HITL** → human help

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**5. Data Flow**

Student Data → Chunk → Embedding → Store
User Query → Embed → Retrieve → LLM → Answer

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**6. Technology Choices**

- ChromaDB → fast retrieval
- LangGraph → workflow control
- LLM → intelligent answers

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**7. Scalability**

- Large student data handle
- Fast query response
- Multi-user support